

A Literature-Implied Negative Answer to the Arbitrary-Spectrum Sampling Problem for $q > 2$

Math discovery packet

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1 Status

This is a lightweight *literature-implied answer*, not a new proof. The source paper is

V. N. Temlyakov, *Sampling discretization of integral norms of the hyperbolic cross polynomials*, arXiv:2005.05967

In Section 4, Open problem 3 asks whether, for every $q \in (2, \infty)$ and dimension d , there are constants $C_i = C_i(q, d)$ and $c(q, d)$ such that for every finite $Q \subset \mathbb{Z}^d$,

$$\mathcal{T}(Q) \in \mathcal{M}(m, q, C_2, C_3) \quad \text{whenever} \quad m \geq C_1 |Q| (\log(2|Q|))^{c(q, d)}.$$

The supporting paper is

F. Dai and V. N. Temlyakov, *Sampling discretization of integral norms and its application*, arXiv:2109.09030.

2 Identification

Remark AR2 in arXiv:2109.09030 gives the obstruction. Let $\Lambda_n = \{k_j\}_{j=1}^n \subset \mathbb{Z}$ be lacunary. For

$$\mathcal{T}(\Lambda_n) = \left\{ \sum_{k \in \Lambda_n} c_k e^{ikx} \right\},$$

the paper recalls the lower bound from KKLT: for every fixed $p > 2$ and fixed equivalence constants C_1, C_2 , the condition

$$\mathcal{T}(\Lambda_n) \in \mathcal{M}(m, p, C_1, C_2)$$

implies

$$m \geq c(p, C_1, C_2) n^{p/2}.$$

Now set $p = q > 2$, $d = 1$, and $Q = \Lambda_n$. If Open problem 3 from arXiv:2005.05967 had a positive answer, then for some constants depending only on q one could take

$$m \leq C n (\log(2n))^c$$

and obtain a valid Marcinkiewicz discretization for all large n . But

$$n (\log n)^c = o(n^{q/2}) \quad (q/2 > 1),$$

contradicting the lacunary lower bound. Therefore the arbitrary-spectrum $q > 2$ statement is false already in dimension one.

3 Scope

This answers only Open problem 3 from the source paper. It does not answer the near-linear hyperbolic-cross question in Open problem 1, where the frequency sets have much more structure, and it does not settle Open problem 2 about the logarithmic entropy exponent.

4 References

- V. N. Temlyakov, *Sampling discretization of integral norms of the hyperbolic cross polynomials*, arXiv:2005.05967.
- F. Dai and V. N. Temlyakov, *Sampling discretization of integral norms and its application*, arXiv:2109.09030.
- The lower bound cited in arXiv:2109.09030 as KKLT, Section D.20.